

Machine Learning Results and Interpretation

Introduction

The objective of the machine learning stage is not only to apply predictive models, but also to evaluate whether the statistical relationships identified during the exploratory data analysis can be transformed into meaningful forecasting signals.

The prediction problem was formulated as a binary classification task where the target variable represents the direction of the next-day portfolio return. The model predicts whether the portfolio return will be positive or negative on the following trading day.

This formulation reflects a realistic financial decision-making scenario because investors are often more interested in predicting market direction than the exact magnitude of returns.

Dataset Preparation

The dataset was divided into training and testing subsets using a chronological time-based split.

- Training Set: 1239 observations
- Test Set: 310 observations

Unlike random sampling, a time-based split preserves the natural sequence of financial data and prevents look-ahead bias. This approach provides a more realistic evaluation framework because models are trained using historical observations and tested on future periods.

Feature Selection

The selected features were directly motivated by the findings obtained during the exploratory data analysis.

The feature set includes:

- BIST100 Returns
- Lagged Portfolio Returns
- Rolling Volatility
- Rolling Moving Averages
- USD/TRY Exchange Rate Returns
- S&P500 Returns
- Market Trend Indicators

These variables were selected because the EDA stage revealed strong market dependence, volatility clustering, heavy-tailed return distributions, and time-varying risk behavior.

The purpose of the feature engineering stage was to capture both market-wide influences and short-term historical patterns that may contain predictive information.

Machine Learning Models

Three different machine learning algorithms were implemented to evaluate different levels of model complexity.

Logistic Regression

Logistic Regression was selected as the baseline model. It assumes a linear relationship between explanatory variables and the probability of a positive return.

Although relatively simple, Logistic Regression provides an important benchmark because it allows evaluation of whether the prediction problem contains a clear linear structure.

Random Forest

Random Forest was selected as the primary ensemble learning model.

The model combines multiple decision trees and can capture:

- Non-linear relationships
- Feature interactions
- Complex market behavior
- Threshold effects

Because financial markets often exhibit non-linear dynamics, Random Forest was expected to outperform the linear baseline model.

Gradient Boosting

Gradient Boosting was included as a more advanced ensemble learning technique.

Unlike Random Forest, Gradient Boosting constructs trees sequentially and attempts to reduce previous prediction errors during each iteration.

Although Gradient Boosting frequently performs well in structured prediction problems, financial markets often contain significant noise which can reduce its effectiveness.

Model Performance Results

Figure X presents the predictive performance of the three machine learning models.

Model	Accuracy	ROC-AUC
Logistic Regression	0.500	0.494
Random Forest	0.516	0.504
Gradient Boosting	0.464	0.472

The results indicate that all models struggled to generate strong predictive performance.

Random Forest achieved the highest accuracy and ROC-AUC scores among all evaluated models. However, the improvement remained extremely limited. The ROC-AUC value remained close to 0.50, suggesting performance only marginally better than random classification.

Logistic Regression produced results almost identical to random guessing. This suggests that future return direction cannot be explained using a simple linear relationship between the selected predictors and the target variable.

Gradient Boosting produced the weakest performance among the three algorithms. Despite its ability to model complex interactions, the model appears to have fitted noise rather than meaningful predictive signals.

Overall, the comparison demonstrates that increasing model complexity does not necessarily improve forecasting performance in financial markets.

Classification Report Interpretation

The classification reports provide additional insight into model behavior beyond simple accuracy measurements.

For the Random Forest model, recall values for positive return observations were slightly higher than those for negative observations. This indicates that the model was somewhat more successful in identifying positive return days.

However, precision values remained relatively modest. Consequently, many positive predictions were incorrect, resulting in a substantial number of false positives.

The balance between precision and recall suggests that the model captures weak tendencies within the data but does not generate reliable trading signals.

Confusion Matrix Interpretation

Figure Y presents the confusion matrix for the Random Forest model.

The confusion matrix reveals that the model correctly identifies some positive and negative observations but continues to generate a large number of classification errors.

Specifically, the model frequently:

- Misclassifies negative returns as positive
- Fails to detect a portion of positive return observations
- Produces balanced but weak predictive performance

These findings further support the conclusion that financial return prediction remains a highly challenging problem.

Feature Importance Analysis

Figure Z presents the feature importance values obtained from the Random Forest model.

The results indicate that BIST100-related variables are the most influential predictors within the model. This finding is highly consistent with the results obtained during the exploratory data analysis, where strong correlations between individual stocks and the market benchmark were identified.

Lagged returns and rolling volatility indicators also contributed to prediction performance. Their presence suggests that recent market conditions contain a limited amount of useful information regarding future market movements.

However, the predictive contribution of these variables remains relatively weak.

USD/TRY and S&P500 variables generally exhibited lower importance values. Although these factors influence financial markets through macroeconomic channels, their direct contribution to short-term return prediction appears limited.

The feature importance analysis reinforces one of the central findings of this project: overall market conditions dominate stock behavior, yet even the most influential predictors possess limited forecasting power.

Connection Between EDA and ML Results

One of the most important findings emerges when machine learning results are compared with the exploratory data analysis.

The EDA stage revealed:

- Strong correlation between stocks and BIST100
- Significant market dependence
- Volatility clustering
- Heavy-tailed return distributions
- Dynamic risk behavior

However, the machine learning stage demonstrated that these relationships do not translate into strong predictive power.

This distinction highlights an important concept in financial analysis:

Correlation does not imply predictability.

Stocks may move together under similar market conditions, but historical information alone is insufficient to reliably predict future return direction.

Overall Machine Learning Interpretation

The machine learning analysis demonstrates that predictive signals exist within the dataset, but these signals are extremely weak.

Although Random Forest achieved the best overall performance, none of the evaluated models produced results substantially better than random classification.

The low accuracy and ROC-AUC values indicate that a large portion of return variability is driven by factors not directly captured within the dataset, including:

- Unexpected news events
- Investor psychology
- Political developments
- Macroeconomic shocks
- Global market uncertainty

Consequently, machine learning models in this context are more useful for understanding market structure and evaluating predictive limitations than for generating highly accurate forecasts.

Conclusion of Machine Learning Stage

The machine learning findings are highly consistent with the results obtained during the exploratory data analysis.

The project demonstrates that:

- Turkish large-cap stocks are strongly influenced by BIST100 market behavior.
- Return distributions are heavy-tailed and non-normal.
- Volatility changes dynamically through time.
- Strong correlations do not guarantee predictive power.
- Machine learning models provide limited short-term forecasting capability.

Overall, the machine learning stage highlights both the strengths and limitations of modern predictive analytics in financial markets. While machine learning techniques successfully identify statistical relationships, the highly dynamic and uncertain nature of financial markets prevents these relationships from being transformed into reliable forecasting signals.